Phase 1: AI Project Proposal
Campus Food Chatbot

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Project topic: Inefficient daily meal selection due to fragmented campus dining data and complex personal constraints
Target users/stakeholders: University students and staff

1 Problem Definition & Scope

Currently, the campus dining resources are fragmented and unevenly digitized, with several canteens continuing to utilize physical menus and lacking a platform that allows for real-time access to campus dining menus. Moreover, the campus community encompasses diverse economic, cultural, and religious backgrounds. When choosing meals, students and staff weigh multiple factors based on their needs, such as budget, distance to the canteen, cuisine style, dietary preferences, religious restrictions, and food allergies, which poses a decision-making challenge. When facing these complex personal considerations, traditional manual searching is time-consuming and inefficient, often leading to unnecessary expenditure of effort. To address this problem, this project aims to introduce an artificial intelligence approach to build an efficient meal recommendation system. The scope of this project includes a hybrid recommendation chatbot that uses an LLM to process natural language, extract the user intent, and perform data enrichment. It integrates this language processing with a rule-based scoring backend using structured menu databases to provide explainable recommendations. The scope excludes restaurant foot traffic tracking or training a customized LLM from scratch.

2 AI Task Formulation

The collectible inputs include the locations, opening hours, and menus of campus canteens and restaurants, where dishes are labeled with ingredients, potential allergens, prices, whether they are cold or hot, and the dish name. Emotion/intent examples along with their corresponding food types, may also be collected. The expected outputs are a table containing a ranked list of the top k recommended dishes with a natural-language explanation for each recommendation, and the details of the recommended dishes. The unit of prediction/decision is a table of ranked recommendations and corresponding details per user query.

Constraints are that recommendations must be limited to currently available menu items and must not ignore or violate allergies, religious or dietary restrictions, budget, or canteen hours. User satisfaction will be measured via a satisfaction survey after each interaction, asking users to rate the relevance and usefulness of the recommendations on a 1-5 scale. Success is defined as achieving a mean score of at least 3.5/5.0. Data sources include canteen and restaurant information and menus obtained via official websites or APIs; public weather APIs. Privacy, consent, and access considerations include obtaining authorization to access menu data and collecting anonymized user data with opt-in consent.

3 Proposed AI Approaches

Present multiple plausible AI approaches. For each approach, briefly explain the core method, required data/resources, expected workflow, and practical feasibility. Divergent solutions are encouraged; do not present only one method under different names.

[Replace this instruction with your proposal content.]

4 Comparative Analysis

Compare the proposed approaches using relevant dimensions such as expected effectiveness, data needs, implementation effort, compute cost, interpretability, robustness, privacy, and deployment constraints. Clearly identify the preferred approach and justify the choice.

A compact comparison table is encouraged when it saves space.

[Replace this instruction with your proposal content.]

5 Expected Impact & Project Feasibility

The system is designed to deliver practical benefits to the campus community by providing individualized food options. It analyses different dimensions such as location, and mood along with historical favorites that can help user to reduce the decision time on choosing the cuisine and enhance the dining experience on campus. It is of paramount importance to prioritize food safety by filtering out any items on the menu that students dislike or that have other restrictions to reduce food-related incidents. Furthermore, this preference data can help campus dining services to adjust their menus and operations in timely manner.

To ensure the feasibility of the project, basic data collection is required, which involves campus menus, different canteen locations and their operating times by searching the internet and public API. Moreover, large amounts of personal preference data also need to be collected through questionnaires to customize the personalization chatbot. Those data will be stored in the PostgreSQL database. For the implementation phase, the system will integrate the LLM API, which is designed for understanding and generating language, with a Python backend to do constraint filtering and ranking. The primary risk is LLM hallucination, where the model may generate some inaccurate recommendations. To mitigate this, the system will validate outputs against the database to make sure the expected output is only menu items.

References

[Author(s)], "[Title]," *[Venue or Source]*, [Year].

[Author(s)], "[Title]," *[Venue or Source]*, [Year].

[Author(s)], "[Title]," *[Venue or Source]*, [Year].